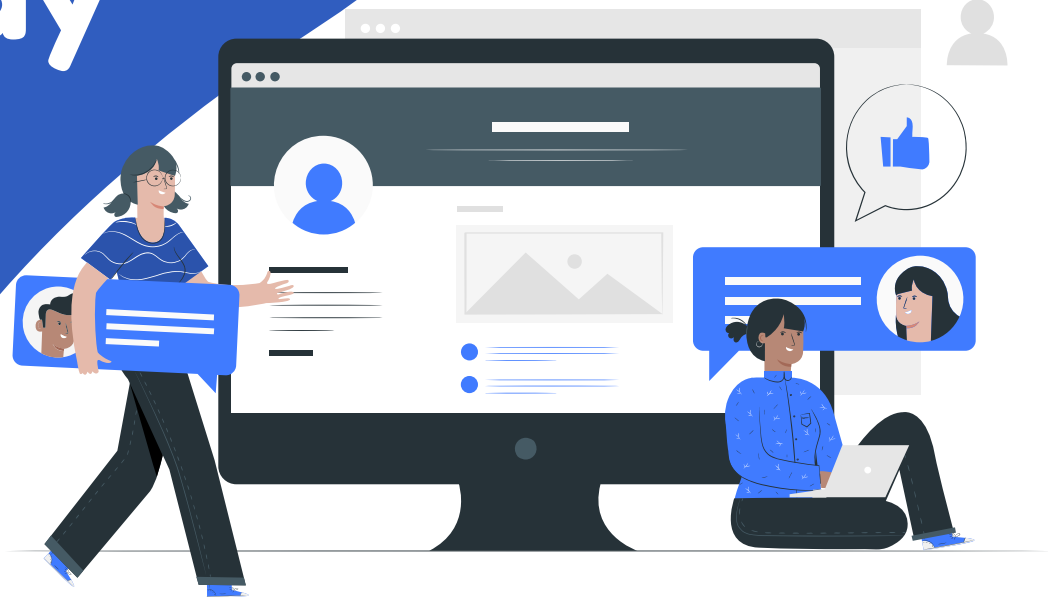


StudyBuddy

Kelompok 1



Background & Problem

Mahasiswa sering mengalami kesulitan dalam mencari:

- Sulit mencari partner akademik yang kompatibel
- Sulit mengelola project collaboration
- Tidak ada centralized academic networking platform
- Tidak ada intelligent recommendation system
- Academic ecosystem masih fragmented

Masalah pada metode konvensional

- Grup chat terlalu umum
- Tidak ada filtering skill atau interest
- Sulit mencari koneksi terpercaya
- Tidak ada sistem recommendation



Dampaknya:

- Kolaborasi menjadi tidak efektif
- Sulit membangun network akademik
- Banyak project team mismatch

Real-World Case Study



LinkedIn menggunakan Graph Database untuk:

- People You May Know
- Mutual connection recommendation
- Job recommendation
- Skill-based networking
- Economic Graph

Problem

Relational Database membutuhkan:

- banyak JOIN
- recursive query
- traversal kompleks

Solution

Menggunakan Graph Database karena:

- Relationship disimpan langsung
- Traversal jauh lebih cepat
- Cocok untuk social networking

Why Graph Database?

Relational Database:

- JOIN query sangat kompleks
- Traversal relationship lambat
- Sulit untuk recommendation system

Neo4j:

- Relationship sebagai first-class citizen
- Traversal antar node sangat cepat
- Natural untuk social graph
- Sangat cocok untuk recommendation engine

Relational DB	Graph DB
Heavy JOIN	Direct Traversal
Complex recursion/query	Flexible relationship
Slow recommendation	Fast recommendation





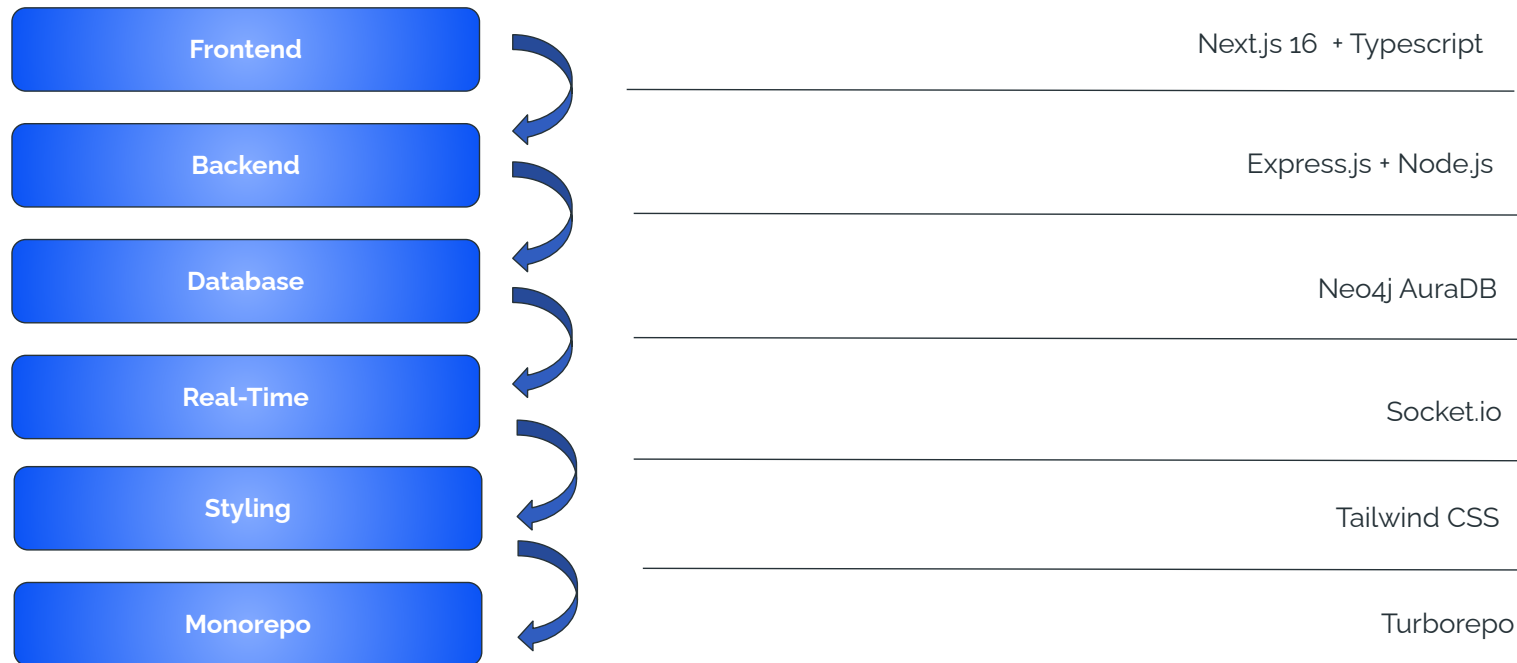
Academic graph collaboration platform berbasis Neo4j Graph Database yang membantu mahasiswa membangun koneksi akademik, mencari partner project, dan mengelola collaborative ecosystem.



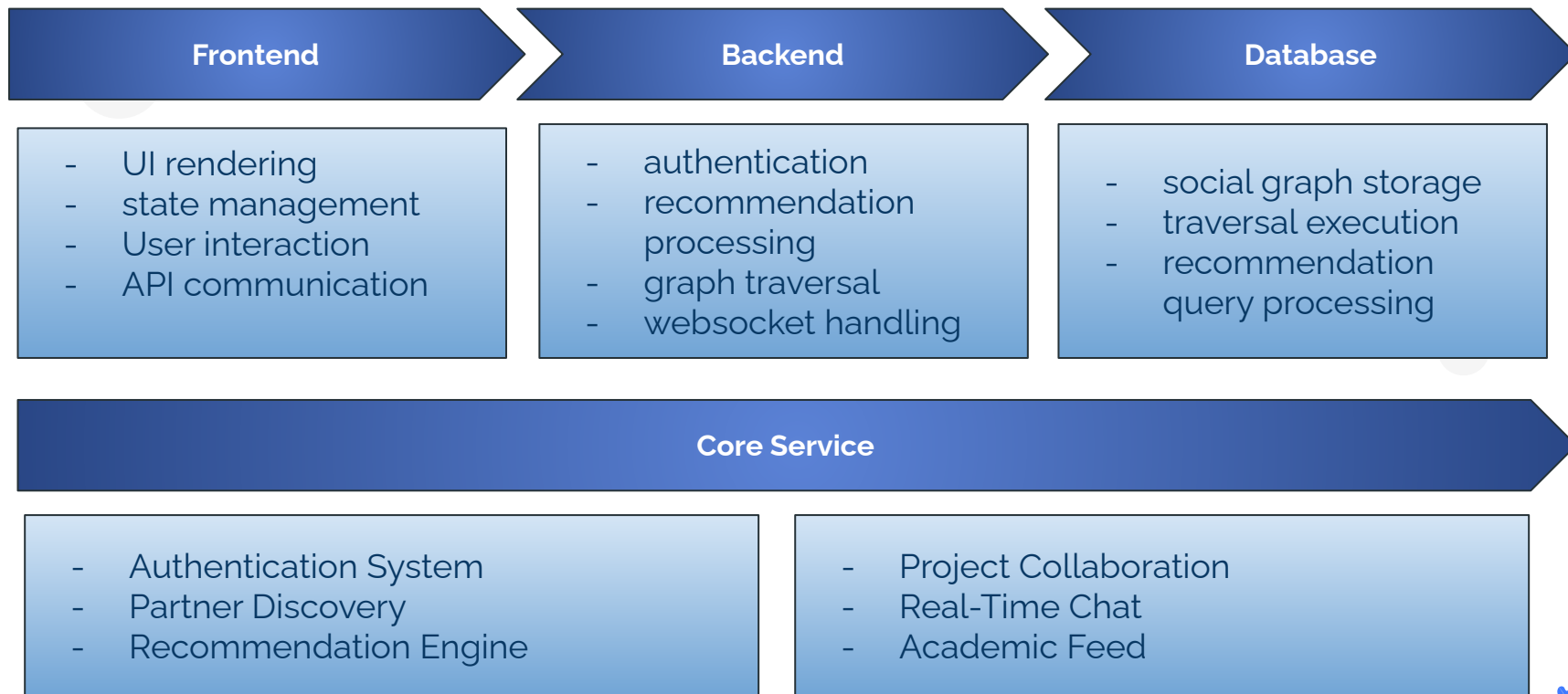
Main Features

- Academic Social Feed
- Partner Discovery System
- Project Collaboration Hub
- Real-time Messaging
- Opportunity Board
- Academic Recommendation Engine

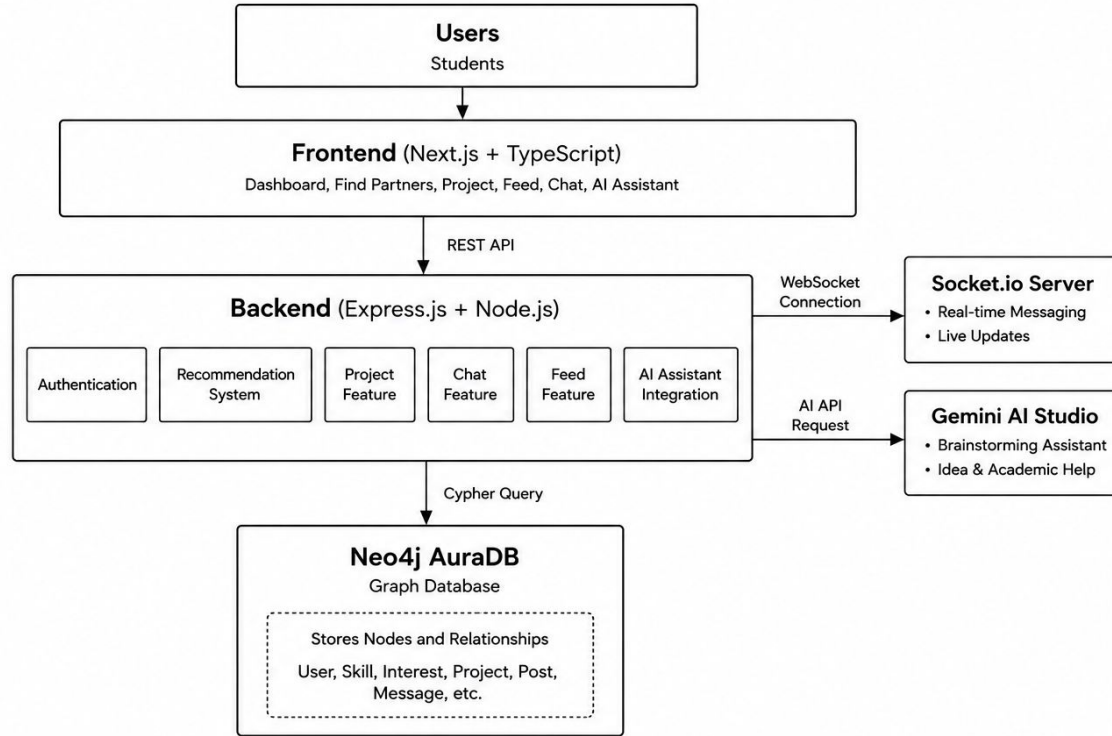
Tech Stack



System Architecture



System Architecture Diagram



Graph Database Model

Main Nodes

- **User**
- Fakultas
- Jurusan
- Angkatan
- **Skill**
- **Interest**
- Mata Kuliah
- **Project**
- Message
- Flashcard
- Course Grade
- Notification
- Opportunity
- **Post**
- Certificate

Main Relationships

- BELONGS_TO_FAKULTAS
- MAJORS_IN
- CLASS_OF
- HAS_SKILL
- INTERESTED_IN
- ENROLLED_IN
- IS_FRIENDS_WITH
- CREATED_PROJECT
- JOINED_PROJECT
- USES_SKILL
- SENT_MESSAGE
- POSTED
- LIKED
- COMMENTED

Graph Relationship Structure

(User)-[:HAS_SKILL]->(Skill)

(User)-[:INTERESTED_IN]->(Interest)

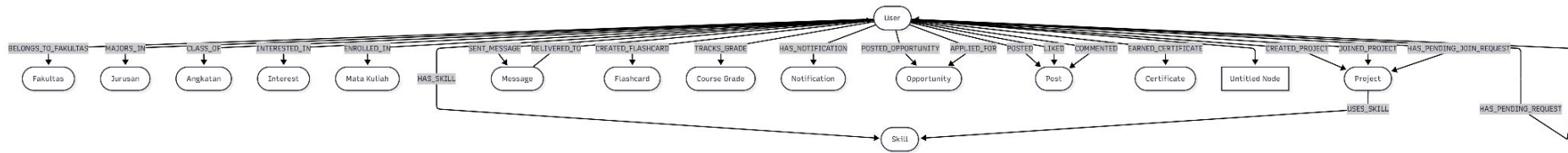
(User)-[:IS_FRIENDS_WITH]->(User)

(User)-[:JOINED_PROJECT]->(Project)

(Project)-[:USES_SKILL]->(Skill)

(User)-[:POSTED]->(Post)

Graph Database Model



Data Modeling

User Properties

- id
- name
- email
- fakultas
- jurusan
- angkatan
- skills
- interests

Project Properties

- title
- description
- required skills



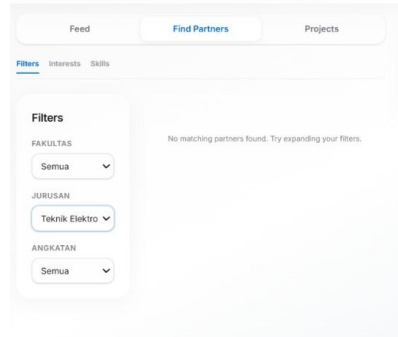
Kenapa Graph Cocok?

Karena seluruh data saling terhubung dan membutuhkan traversal relationship.

Search & Filter System

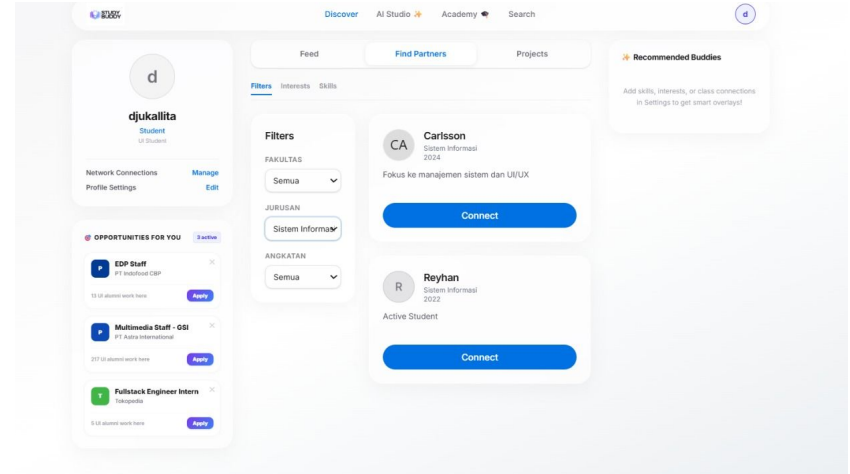
Filters

- Fakultas
- Jurusan
- Angkatan



Workflow

1. User memilih filter
2. Frontend mengirim query parameter
3. Backend menjalankan Cypher query
4. Neo4j mengembalikan hasil sesuai filter



Recommendation System

Recommendation by Interest

Mencari user dengan interest yang sama.

Recommendation by skills

Mencari user dengan skill yang mirip.

Recommendation by social proximity

Menggunakan:

- friendship relationship
- jurusan
- angkatan

Partner Discovery Feature

Features:

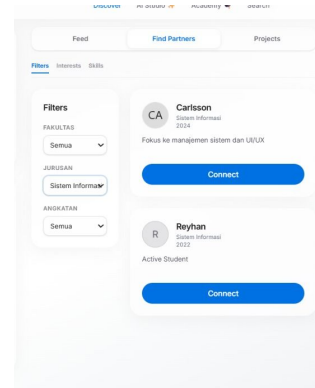
- filter recommendation
- recommendation tabs
- connect button
- discover page

Tabs:

- Filters
- Interests
- Skills

Purpose

Membantu user menemukan partner akademik dengan lebih mudah.



Chat System

Technology

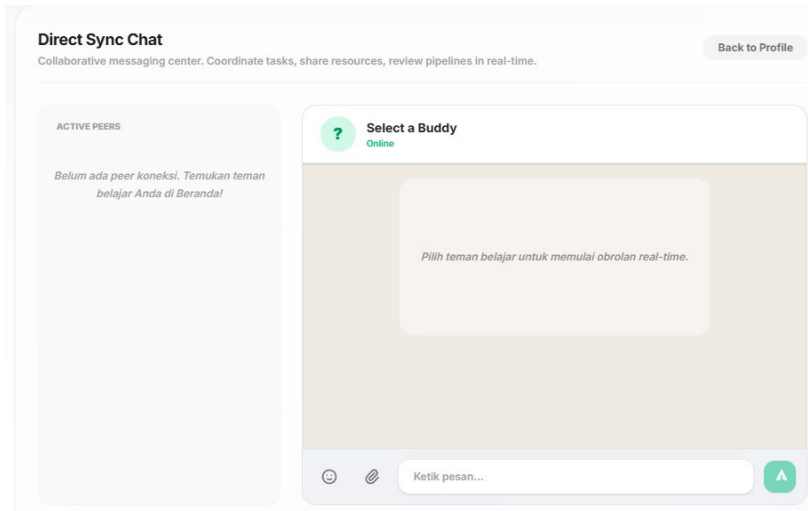
Socket.io

Features

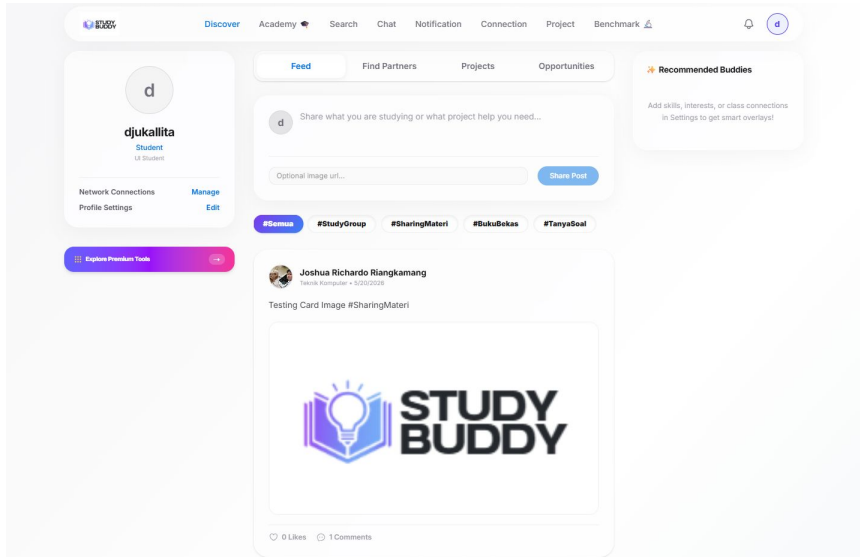
- real-time messaging
- instant communication
- live update

Purpose

Mempermudah komunikasi antar user dan collaborator.



UI/UX Design



Design Style

- modern UI
- clean interface
- glassmorphism layout

Technologies

- Tailwind CSS
- responsive layout

Purpose

Membuat platform nyaman dan efisien digunakan mahasiswa.

Lessons Learned

Technical Lesson

Graph Database sangat cocok untuk social platform

Cypher query powerful untuk relationship traversal

Relationship modeling sangat penting

Team Lesson

Teamwork dan pembagian peran sangat penting

GitHub membantu dalam collaboration

Debugging membutuhkan komunikasi tim

Future Improvements

Planned Features

mobile application

smarter recommendation ranking

better notification system

improved AI assistant

improved collaboration system

Conclusions

StudyBuddy Successfully Implements

- Neo4j Graph Database
- Recommendation System
- Social Relationship Traversal
- Academic Collaboration Platform
- Real-Time Communication



Neo4j Graph Database sangat cocok digunakan untuk platform berbasis relationship dan recommendation system seperti StudyBuddy.

Role Distribution

Team Member	Responsibilities
Mohammad Ariq Haqi	Responsible for the formal design of the graph schema, optimizing node properties, and writing the advanced Cypher queries needed for the matchmaking logic.
Joshua Richardo Riangkamang	Building the server-side application using Express.js to interface with the Neo4j driver and managing the flow of data between the graph and the user.
Vincenzo Fabian Tisila	Developing the user interface with Next.js to provide a seamless experience for students searching for partners and visualizing their academic network.
Djukallita Tafiana Djoewarsa	Ensuring the project adheres to the LinkedIn case study principles (fixing bugs) and managing the documentation, including the technical report and final presentation.
Reyhan Batara	Handling the initial data ingestion, including importing course and student datasets into Neo4j, and testing query performance to ensure the system scales efficiently.

Terima Kasih

